

SwiftPay - International Remittance & Currency Exchange Portal

1. Introduction

SwiftPay is a web-based **International Remittance & Currency Exchange** platform enabling customers to **send and receive cross-border payments**, manage **beneficiaries**, obtain **real-time FX quotes**, comply with **KYC/AML** checks, and **track remittance status** end-to-end.

It supports **account-to-account**, **cash pickup** (reference), and **mobile wallet** (reference) payouts, with **corridor rules**, **fees & margin configuration**, **rate locks**, and **document collection** for regulatory reporting.

The solution uses a **REST API-based backend** and **Angular/React** frontend. Phase-1 treats **FX rate feeds**, **correspondent rails**, and **payout networks as references** (simulated connectors).

Actors / Users

- **Retail Customer:** Creates remittance requests, obtains quotes, uploads documents, tracks status.
- **Branch/Agent:** Assists customers in-store, performs KYC capture, initiates transactions.
- **Compliance Analyst:** Reviews KYC/AML flags, approves/holds transactions.
- **Operations Officer:** Manages queues, reconciles settlements, handles amendments/cancellations.
- **Treasury/FX Dealer:** Manages rate margins, corridor spreads, and rate lock policies.
- **Admin:** Configures corridors, fees, limits, payout modes, and reports.

2. Module Overview

2.1 Identity, KYC & Access Management

Authentication, RBAC, KYC capture, audit logs.

2.2 Customer Profile & Beneficiaries

Customer records, verification status, beneficiary master per corridor.

2.3 FX Quotes, Fees & Rate Lock

Live FX quotes (reference), margin application, fees, and rate lock windows.

2.4 Remittance Initiation & Validation

Corridor checks, limits/velocity, purpose of remittance, document requirements.

2.5 Compliance Screening & Approvals

KYC/AML/PEP/sanctions checks, holds, approvals, enhanced due diligence.

2.6 Routing & Payout Instruction (Reference)

Select partner/correspondent, construct payout payload (reference), track acks.

2.7 Settlement, Funding & Reconciliation (Reference)

Settlement batches, nostro funding references, payout reconciliation.

2.8 Amendments, Cancellations & Refunds

Pre-payout amendment/cancel; refunds post-cancellation (reference).

2.9 Reporting & Regulatory

Remittance registers, corridor volumes, exception KPIs, regulatory packs (reference).

2.10 Notifications & Alerts

Status updates, KYC/AML requests, rate expiry, payout confirmation.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST microservices (Identity/KYC, Profiles, Quotes, Remittance, Compliance, Routing, Settlement, Reports)
- **Database:** Relational Database (MySQL/PostgreSQL/SQL Server)
- **Integrations (Phase-1):** FX rate feed, correspondent rails, payout networks, and payment rails are **simulated as references** (payloads stored, not transmitted)

4. Module-Wise Design

4.1 Identity, KYC & Access Management Module

Features:

- Authentication, RBAC, session control, KYC capture and statuses, immutable audit logs

Entities:

- **User**
 - UserID
 - Name
 - Role (Customer/Agent/Compliance/Ops/Treasury/Admin)
 - Email
 - Phone
 - Status (Active/Locked/Disabled)

- **KYCRecord**
 - KYCID
 - UserID
 - KYCLevel (Min/Full/Enhanced)
 - VerificationStatus (Pending/Verified/Rejected)
 - VerifiedDate
 - Notes
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - Resource
 - Timestamp

4.2 Customer Profile & Beneficiaries Module

Features:

- Maintain customer data and beneficiary details per corridor with validation

Entities:

- **Customer**
 - CustomerID
 - UserID
 - DOB
 - AddressJSON
 - Nationality
 - RiskRating (Low/Med/High)
 - Status
- **Beneficiary**
 - BeneficiaryID
 - CustomerID
 - Name
 - Country

- PayoutMode (Account/CashPickup/MobileWallet)
- BankName
- BankCountry
- AccountOrWalletNo
- IFSC_IBAN_SWIFT
- VerificationStatus (Pending/Verified/Rejected)
- AddedDate
- Status

4.3 FX Quotes, Fees & Rate Lock Module

Features:

- Get FX quote (reference feed), apply corridor margin & fees, lock rate within validity

Entities:

- **FXQuote**
 - QuoteID
 - FromCurrency
 - ToCurrency
 - MidRate
 - MarginBps
 - OfferedRate
 - QuoteTime
 - ValidUntil
 - Status (Active/Expired)
- **FeeRule**
 - FeeRuleID
 - Corridor (FromCountry→ToCountry)
 - PayoutMode
 - FeeType (Flat/Percent/Tiered)
 - FeeValue
 - MinFee

- MaxFee
- EffectiveFrom
- EffectiveTo
- Status
- **RateLock**
 - LockID
 - QuoteID
 - CustomerID
 - LockStart
 - LockExpiry
 - Status (Locked/Released/Expired)

4.4 Remittance Initiation & Validation Module

Features:

- Create remittance requests, validate eligibility, compute fees, capture purpose & docs

Entities:

- **RemittanceRequest**
 - RemitID
 - CustomerID
 - BeneficiaryID
 - FromCurrency
 - ToCurrency
 - SendAmount
 - ReceiverAmount
 - QuoteID
 - FeeApplied
 - RateApplied
 - PurposeCode
 - SourceOfFunds
 - CreatedDate

- Status
(Draft/Validated/ComplianceHold/Routing/Queued/Paid/Cancelled/Refunded)
- **RemitValidation**
 - ValidationID
 - RemitID
 - RuleName (Limit/Velocity/Docs/Corridor)
 - Result (Pass/Fail)
 - Message
 - CheckedDate
- **Document**
 - DocumentID
 - RemitID
 - DocType (IDProof/SoF/Invoice/Declaration)
 - FileURI
 - UploadedDate
 - VerificationStatus (Pending/Approved/Rejected)

4.5 Compliance Screening & Approvals Module

Features:

- Sanctions/PEP/AML checks (reference), risk scoring, manual review/approval holds

Entities:

- **ComplianceCheck**
 - CheckID
 - RemitID
 - CheckType (Sanctions/PEP/AML/Geo)
 - Result (Clear/Flag/Hold)
 - Severity (Low/Medium/High)
 - CheckedDate
- **ComplianceDecision**
 - DecisionID

- RemitID
- AnalystID
- Decision (Approve/Hold/Reject)
- Notes
- DecisionDate

4.6 Routing & Payout Instruction Module (Reference)

Features:

- Select corridor partner/correspondent, create payout instruction payload (reference), store ack refs and statuses

Entities:

- **RoutingRule**
 - RuleID
 - Corridor
 - PayoutMode
 - PartnerCode
 - Priority
 - Status
- **PayoutInstruction**
 - InstructionID
 - RemitID
 - PartnerCode
 - PayloadJSON
 - SentDate
 - AckRef
 - PartnerStatus (Sent/Ack/Rejected/Settled)

4.7 Settlement, Funding & Reconciliation Module (Reference)

Features:

- Aggregate remittances into settlement batches; nostro funding ref; basic recon of partner acks vs internal

Entities:

- **SettlementBatch**
 - BatchID
 - Corridor
 - PeriodStart
 - PeriodEnd
 - ItemCount
 - TotalSendAmount
 - TotalPayoutAmount
 - CreatedDate
 - Status (Open/Posted/Reconciled)
- **ReconciliationRecord**
 - ReconID
 - ReferenceType (Remit/Instruction/PartnerAck)
 - ReferenceID
 - ExpectedAmount
 - ActualAmount
 - Result (Matched/Mismatched)
 - ReconDate
 - Notes

4.8 Amendments, Cancellations & Refunds Module**Features:**

- Allow pre-payout amendment/cancellation; record refunds after cancellation (reference)

Entities:

- **Amendment**
 - AmendmentID
 - RemitID
 - FieldChanged
 - OldValue

- NewValue
- RequestedBy
- RequestedDate
- Status (Pending/Approved/Rejected)
- **Cancellation**
 - CancellationID
 - RemitID
 - Reason
 - RequestedDate
 - Status (Requested/Approved/Posted)
- **RefundRef**
 - RefundID
 - RemitID
 - Amount
 - RefundDate
 - Method (Original/Fallback)
 - Status (Initiated/Completed/Failed)

4.9 Reporting & Regulatory Module

Features:

- Operational MIS and regulatory reference packs (LRS tracking, corridors, limits)

Entities:

- **RemitReport**
 - ReportID
 - Scope (Corridor/Period/Customer)
 - Metrics (TxnCount, Volume, AvgTicket, Fees, RejectRate)
 - GeneratedDate

4.10 Notifications & Alerts Module

Features:

- In-app notifications for status changes, document requests, rate expiry, payout confirmation

Entities:

- **Notification**
 - NotificationID
 - UserID
 - RemitID
 - Message
 - Category (KYC/Quote/Compliance/Routing/Payout/Refund)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React UI, Spring Boot/.NET Core backend, local RDBMS
- **Production:** Cloud/on-prem, API gateway, WAF, stateless microservices, schedulers for **rate expiry**, **settlement batches**, **recon packs**; centralized logging & metrics

6. Database Design

Tables:

- User(UserID, Name, Role, Email, Phone, Status)
- KYCRecord(KYCID, UserID, KYCLevel, VerificationStatus, VerifiedDate, Notes)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)
- Customer(CustomerID, UserID, DOB, AddressJSON, Nationality, RiskRating, Status)
- Beneficiary(BeneficiaryID, CustomerID, Name, Country, PayoutMode, BankName, BankCountry, AccountOrWalletNo, IFSC_IBAN_SWIFT, VerificationStatus, AddedDate, Status)
- FXQuote(QuoteID, FromCurrency, ToCurrency, MidRate, MarginBps, OfferedRate, QuoteTime, ValidUntil, Status)
- FeeRule(FeeRuleID, Corridor, PayoutMode, FeeType, FeeValue, MinFee, MaxFee, EffectiveFrom, EffectiveTo, Status)
- RateLock(LockID, QuoteID, CustomerID, LockStart, LockExpiry, Status)

- RemittanceRequest(RemitID, CustomerID, BeneficiaryID, FromCurrency, ToCurrency, SendAmount, ReceiverAmount, QuoteID, FeeApplied, RateApplied, PurposeCode, SourceOfFunds, CreatedDate, Status)
- RemitValidation(ValidationID, RemitID, RuleName, Result, Message, CheckedDate)
- Document(DocumentID, RemitID, DocType, FileURI, UploadedDate, VerificationStatus)
- ComplianceCheck(CheckID, RemitID, CheckType, Result, Severity, CheckedDate)
- ComplianceDecision(DecisionID, RemitID, AnalystID, Decision, Notes, DecisionDate)
- RoutingRule(RuleID, Corridor, PayoutMode, PartnerCode, Priority, Status)
- PayoutInstruction(InstructionID, RemitID, PartnerCode, PayloadJSON, SentDate, AckRef, PartnerStatus)
- SettlementBatch(BatchID, Corridor, PeriodStart, PeriodEnd, ItemCount, TotalSendAmount, TotalPayoutAmount, CreatedDate, Status)
- ReconciliationRecord(ReconID, ReferenceType, ReferenceID, ExpectedAmount, ActualAmount, Result, ReconDate, Notes)
- Amendment(AmendmentID, RemitID, FieldChanged, OldValue, NewValue, RequestedBy, RequestedDate, Status)
- Cancellation(CancellationID, RemitID, Reason, RequestedDate, Status)
- RefundRef(RefundID, RemitID, Amount, RefundDate, Method, Status)
- RemitReport(ReportID, Scope, Metrics, GeneratedDate)
- Notification(NotificationID, UserID, RemitID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Customer Portal:** Create remittance, get quote, lock rate, upload docs, track status, view history.
- **Agent Console:** Assisted onboarding, KYC capture, initiate transactions, print receipts.
- **Compliance Workbench:** Screening results, flags, holds/approvals, evidence review.
- **Operations Dashboard:** Queues (Validation/Compliance/Routing), settlement batches, recon mismatches.
- **Treasury/FX Console:** Margin settings, corridor spreads, rate lock policies.

- **Admin Panel:** Corridor setup, payout mode rules, fees, limits, partners, reports.

8. Non-Functional Requirements

- **Performance:** Handle **50,000 concurrent users**; sub-second quote fetch (from cache in Phase-1), swift status transitions.
- **Security:** PII encryption at rest & in transit, RBAC, immutable audit; data minimization for documents.
- **Scalability:** Horizontal scaling for **quote**, **remittance**, and **routing** services; partition by corridor/region.
- **Availability:** **99.9% uptime**; graceful degradation (analytics/report packs deferred).
- **Maintainability:** Modular microservices; policy & rule configs; API versioning; automated schema migrations.
- **Observability:** Central logs; KPIs (conversion, average fee, corridor latency, compliance hold rate); alerting.

9. Assumptions & Constraints

- Phase-1 **does not integrate live** FX feeds/correspondent/payout APIs—**payloads and acks are stored as references**.
- Payment collection and disbursement rails are **out of scope** (captured as references only).
- Regulatory packs (e.g., outward remittance registers, LRS caps) are **reference summaries**; no regulator gateway integration.
- In-app notifications only; no email/SMS in Phase-1.
- Fully implementable using **Java Spring Boot/ASP.NET Core + Angular/React** and a relational database.